Frozen-Encoder Guidance for Diffusion Text Generation on a Semantic Map

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This implementation steers a frozen local diffusion language model toward a clicked point on a fixed embedding map. It adds encoder-input gradients inside denoising while retaining the outer generate-and-verify search. Real probes confirm that guidance changes native sampling, decoded text, and its measured location. They do not establish a general quality or efficiency advantage.

1 Target and native runtime

The map contains 100 authored technical paragraphs. A pinned all-MiniLM-L6-v2 encoder [1] produces normalized 384-dimensional vectors; ordinary unwhitened PCA supplies a fixed mean mu and two orthonormal directions V. Canonical scoring uses the normal tokenizer, attention-mask mean pooling, float32 CPU execution, and L2 normalization. Inputs exceeding 256 encoder tokens, including special tokens, are rejected.

$$\pi(e) = V^T(e - \mu), \qquad d(x, y^*) = \|\pi(E(x)) - y^*\|_2.$$

The generator is the four-bit mlx-community/diffusiongemma-26B-A4B-it-4bit checkpoint, pinned to revision a7a81407613811e8ba63af92ac0d852b809e191f. DiffusionGemma [2] runs through MLX-VLM 0.7.11 [3]. Each trajectory uses a 128-token canvas, vocabulary size 262,144, and a 32-pass maximum. A native prompt KV cache supplies conditioning throughout that trajectory.

The entropy-bound sampler can revise the whole canvas at each pass. Its confidence/stability rule can stop before the pass budget. The hook receives detached, temperature-adjusted native logits before sampling. Its returned logits drive sampling, entropy-based acceptance, self-conditioning, and stopping decisions. Consequently, modifying logits can alter both token values and the subsequent trajectory. No backward graph passes through the generator.

2 Sparse encoder surrogate

At a guided pass, the hook selects 32 vocabulary entries per position on the native device. Only compact logits, IDs, masks, and retained probability mass cross to CPU. A token bridge decodes generator tokens in isolation, re-tokenizes their surfaces with MiniLM, and averages the resulting encoder input vectors. Unsupported or empty surfaces are omitted; mixtures are renormalized over represented surfaces. Encoder CLS/SEP tokens, attention masks, mean pooling, and output normalization remain explicit.

This bridge is approximate: isolated surfaces discard contextual segmentation, multiple encoder tokens collapse into one position, and top-k selection removes alternatives. Final EOS clipping and ordinary re-tokenization can also change the scored input. The surrogate is not the canonical text encoder evaluated on a differentiable string.

On fixed selected supports, q_i is the conditional softmax distribution and q_i^0 is the original native proposal. For editable positions M, the loss is:

$$L(z) = \|\pi(E_{\text{soft}}(z)) - y^*\|_2^2 + \frac{\lambda}{|M|} \sum_{i \in M} \mathrm{KL}(q_i \| q_i^0).$$

The update count is configurable from 1 to 16 per pass (default two), retaining the same reference proposal. The first KL gradient is zero because q_i initially equals q_i^0; the penalty becomes active after movement. The compact gradient is globally L2-normalized, direction-clipped, and scaled. A trust bound clamps cumulative per-logit changes relative to the original proposal. Fixed positions remain unchanged. Zero strength returns the original native tensor.

3 Outer search and hard verification

The outer loop remains sequential best-of-N generation. Four nearest corpus paragraphs provide the same context for each candidate; seeds differ. Within a trajectory, selected passes additionally perform the encoder-guided updates:

```
for each candidate seed:
  prefill native prompt cache; initialize canvas
  for each executed denoising pass:
    z0 <- frozen generator logits
    z <- m sparse encoder-loss updates if scheduled
         otherwise z0
    advance native sampler using z; stop if converged
  x <- decode final tokens
  skip empty or duplicate x
  e <- canonical_encode_text(x)
  publish (x, pi(e), distance(pi(e), target))
return the published text with smallest hard distance
```

Only normally re-tokenized final text is ranked. Traces retain losses, applied logit changes, executed passes, and hard coordinates. Cancellation preserves verified candidates.

4 What the probes establish

On an M4 Max with 128 GiB memory, paired probes used the same corpus target, neighborhood context, seed 42, checkpoint, and native generation budget.

Profile	Canonical distance	Observed intervention
Unguided	0.07483	Nine native passes
Strong late guidance	0.09050	Four guided passes; text changed
Strong early guidance	0.02411	Thirteen guided passes; text changed

Early guidance ran at all 13 executed passes, with two encoder updates each and a maximum native logit change of approximately four. Late guidance used a tighter bound and began at pass six. These are different intervention profiles on one development target, not a held-out benefit benchmark. The worsening case demonstrates why reducing surrogate loss cannot substitute for canonical verification. No blind factuality or coherence assessment was performed.

5 Computational cost and next gate

Candidates run sequentially, each requiring full native generator passes. Top-k sparsity reduces the reward bridge, not the generator vocabulary computation. The strong early profile adds 26 encoder forward/backward pairs, bridging, transfers, and synchronization. Increasing inner updates adds encoder work, not generator passes. Prompt caching operates within each trajectory; prefill is repeated across candidates. Decoder graph compilation is disabled with compile_graph=False.

Cold/warm timing differences do not demonstrate acceleration. The next meaningful gate is an equal end-to-end wall-clock comparison against spending the same budget on more unguided diffusion samples, across multiple targets. Report hard error together with coherence, factuality, diversity, memory, and latency. Entropy-aware hard/soft estimators [4] are a separate comparison, not an implemented guarantee. Low two-dimensional distance alone does not establish semantic equivalence or useful text.

References

[1] Sentence Transformers. all-MiniLM-L6-v2. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[2] Google. DiffusionGemma model card. https://ai.google.dev/gemma/docs/diffusiongemma/model_card

[3] MLX-VLM. Native inference implementation; version 0.7.11 used here. https://github.com/Blaizzy/mlx-vlm

[4] Tejaswi et al. Entropy Aware Reward Guidance for Diffusion Language Model Alignment. v2, 2026. https://arxiv.org/abs/2602.05000